Mathematics practice booklet: Grade 9 expectations

Grade 9 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 9 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: A-SSE.A.1, A-SSE.B.3, A-APR.A.1, A-CED.A.1, A-CED.A.4, A-REI.A.1, A-REI.B.3, A-REI.B.4, A-REI.C.6, F-IF.A.2, F-LE.A.1

A worked example

Solve x^2 + 2x - 15 = 0 by factoring. Then check one of your answers.

1. Look for two numbers that multiply to -15 and add to 2: they are 5 and -3.
2. So x^2 + 2x - 15 = (x + 5)(x - 3) = 0.
3. A product is 0 only when one factor is 0: x + 5 = 0 or x - 3 = 0.
4. So x = -5 or x = 3.
5. Check x = 3: 9 + 6 - 15 = 0. It works.

x = -5 or x = 3

Questions

1. Layla buys x tickets at AED 4 each and uses a AED 9 voucher, so she pays AED 19. Solve 4x - 9 = 19 to find how many tickets she buys.

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2. Solve: 3(2x - 1) = 5x + 4

   A. x = 1

   B. x = 5

   C. x = 7

   D. x = -1

3. Solve the inequality: 2x + 5 > 17

   A. x > 12

   B. x > 11

   C. x < 6

   D. x > 6

4. Solve the inequality: -3x <= 12. What happens to the inequality sign, and why?

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5. The perimeter of a rectangle is P = 2l + 2w. Solve the formula for w. Then find w when P = 30 and l = 9.

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6. Solve the system: y = 3x - 2 and 2x + y = 13

   A. x = 7, y = 3

   B. x = 3, y = 7

   C. x = 3, y = 11

   D. x = 2, y = 4

7. Solve the system by elimination: 3x + 2y = 16 and x - 2y = 0

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8. At the souq, 2 bags of dates and 1 jar of honey cost AED 40, and 1 bag of dates and 1 jar of honey cost AED 28. Find the price of a bag of dates and of a jar of honey.

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9. If f(x) = 2x^2 - 3x + 1, find f(3).

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10. Which is the expansion of (x + 3)(x - 5)?

    A. x^2 - 2x - 15

    B. x^2 + 2x - 15

    C. x^2 - 15

    D. x^2 + 8x - 15

11. Solve by factoring: x^2 - 5x + 6 = 0

    A. x = -2 or x = -3

    B. x = 1 or x = 6

    C. x = 2 or x = 3

    D. x = 5 or x = 6

12. Noura has AED 50. Karak costs AED 3 a cup and she also buys a snack for AED 8. Write an inequality for the number of cups c she can buy, and solve it.

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13. Maryam's savings start at AED 200 and grow by AED 50 each month. Her cousin Hamdan starts with AED 100 and his savings double each month. Who has more after 4 months? Which one is linear and which is exponential?

Section 2: Shapes, measuring and space

Standards practised: G-GPE.B.5, G-GPE.B.7, G-GMD.A.3

A worked example

Line A is y = 2x + 3. Write the equation of the line through (4, 1) that is perpendicular to line A.

1. Line A has slope 2.
2. A perpendicular line has the negative reciprocal slope: -1/2.
3. Use y - 1 = -1/2 (x - 4), so y - 1 = -1/2 x + 2.
4. Add 1 to both sides: y = -1/2 x + 3.
5. Check: at x = 4, y = -2 + 3 = 1. The line passes through (4, 1).

y = -1/2 x + 3

Questions

1. Find the distance between (1, 2) and (7, 10).

   A. 10 units

   B. 14 units

   C. 100 units

   D. 8 units

2. Find the midpoint of the segment from (-2, 4) to (6, 10).

   A. (4, 14)

   B. (2, 3)

   C. (2, 7)

   D. (8, 6)

3. On a metro map grid, the midpoint between two stations is at (3, 5). One of the stations is at (1, 2). Where is the other station?

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4. Find the slope of the line through (2, 3) and (6, 11).

   A. 1/2

   B. 2

   C. 4

   D. 8

5. Write the equation of the line through (0, 5) that is parallel to y = 2x + 1.

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6. A line has slope 3. What is the slope of any line perpendicular to it?

   A. 3

   B. -3

   C. 1/3

   D. -1/3

7. Are the lines y = 2/3 x + 1 and y = -3/2 x + 4 perpendicular? Explain.

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8. Write the equation of the line through (1, 4) that is perpendicular to y = -1/2 x + 3.

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9. On a plan, a triangular garden has corners at (0, 0), (4, 0) and (4, 3), in metres. How much fencing goes all the way round it?

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10. A cylindrical water tank has radius 7 m and height 10 m. Find its volume using 22/7 for pi.

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11. A cone has radius 6 cm and height 10 cm. Find its volume using 3.14 for pi.

    A. 1,130.4 cubic cm

    B. 376.8 cubic cm

    C. 113.04 cubic cm

    D. 37.68 cubic cm

12. A ball has radius 3 cm. Find its volume in terms of pi, and then using 3.14. If the radius doubles, how many times as large does the volume become?

Section 3: Data, graphs and chance

Standards practised: S-ID.A.1, S-ID.A.2, S-ID.A.3, S-ID.B.5, S-ID.B.6, S-ID.C.7, S-ID.C.8, S-ID.C.9

A worked example

Find the mean and the standard deviation of the data: 4, 6, 8, 10, 12.

1. Mean: (4 + 6 + 8 + 10 + 12) / 5 = 40 / 5 = 8.
2. Distances from the mean: -4, -2, 0, 2, 4. Square them: 16, 4, 0, 4, 16.
3. Add the squares: 40. Divide by 5: 8. This is the variance.
4. The standard deviation is the square root of 8, about 2.8.

Mean 8; standard deviation about 2.8

Questions

1. A survey of 60 Grade 9 students asked how they get to school. Girls: 12 walk and 18 take the bus. Boys: 15 walk and 15 take the bus. How many students walk in all?

   A. 12

   B. 27

   C. 30

   D. 33

2. In the same survey, what percentage of the girls walk? What percentage of the boys walk?

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3. In the same survey, what percentage of ALL the students are boys who take the bus?

   A. 15%

   B. 50%

   C. 25%

   D. 30%

4. A line of fit for the exam score y against the hours of study x is y = 5x + 50. Predict the score for 4 hours of study.

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5. In y = 5x + 50, explain what the slope 5 and the y-intercept 50 mean.

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6. A line of fit predicts a score of 70 but the real score was 74. Find the residual (actual minus predicted).

   A. 4

   B. -4

   C. 70

   D. 74

7. Which correlation coefficient shows the strongest negative relationship?

   A. r = 0.9

   B. r = 0.1

   C. r = -0.2

   D. r = -0.8

8. Children with bigger shoe sizes tend to read better. Does a bigger shoe size cause better reading? Explain.

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9. Data: 2, 4, 5, 7, 8, 9, 12. What is the interquartile range?

   A. 10

   B. 7

   C. 5

   D. 3

10. For the same data, which five numbers mark the ends and the box of a box plot?

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11. Daily visitors to a small shop: 5, 6, 7, 8, 9. On a festival day 40 visitors came and this number was added to the list. Find the mean and the median before and after. Which is a better description of a usual day after adding it?

Section 4: Numbers and calculating

Standards practised: N-RN.A.1, N-RN.A.2, N-RN.B.3, N-Q.A.1

A worked example

Evaluate 8^(2/3) and then write 5^(1/2) as a radical.

1. The exponent 2/3 means: take the cube root, then square. The cube root of 8 is 2.
2. Square it: 2^2 = 4, so 8^(2/3) = 4.
3. An exponent of 1/2 means the square root, so 5^(1/2) is the square root of 5.

4 and the square root of 5

Questions

1. Which is equal to 2^5 × 2^3?

   A. 2^8

   B. 2^15

   C. 4^8

   D. 2^2

2. Simplify and evaluate: 5^6 / 5^4

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3. Evaluate 16^(3/2).

   A. 24

   B. 4

   C. 64

   D. 8

4. Write the cube root of 5 using a rational exponent.

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5. Evaluate 5^-2.

   A. 1/25

   B. -25

   C. -10

   D. 25

6. Which of these numbers is irrational?

   A. 1/2 + 3/4

   B. 3 + the square root of 2

   C. the square root of 16

   D. 0.125

7. Is the product of the square root of 2 and the square root of 8 rational or irrational? Show why.

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8. A falcon flies at 108 km per hour. Convert this to metres per second.

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9. A family car uses 8 litres of fuel for every 100 km. How many litres does it use on a 140 km trip?

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10. Which is 45,000,000 written in scientific notation?

    A. 4.5 × 10^6

    B. 4.5 × 10^8

    C. 4.5 × 10^7

    D. 4.5 × 10^-7

11. Calculate (2 × 10^4) x (3 × 10^3) and (6 × 10^8) / (3 × 10^5). Give each answer in scientific notation.

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12. A desalination plant makes 3 × 10^6 litres of fresh water a day. How many litres does it make in 7 days? Write the answer in scientific notation, and say which unit you would use to talk about it more easily.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. Add 9 to both sides: 4x = 28. Divide by 4: x = 7 tickets. Check: 4 × 7 - 9 = 19.
2. C. x = 7. Distribute first: 6x - 3 = 5x + 4, so x = 4 + 3 = 7. If your child chose A (x = 1), they subtracted 3 on the right instead of adding it.
3. D. x > 6. Subtract 5: 2x > 12. Divide by 2: x > 6. If your child chose A (x > 12), they forgot to divide by 2.
4. Divide both sides by -3: x >= 12 / (-3) = -4. The sign turns around because we divided by a negative number.
5. Take 2l from both sides: 2w = P - 2l. Divide by 2: w = (P - 2l) / 2. With P = 30 and l = 9: (30 - 18) / 2 = 6.
6. B. x = 3, y = 7. Put y = 3x - 2 into the other equation: 2x + 3x - 2 = 13, so 5x = 15, x = 3 and y = 7. If your child chose A (x = 7, y = 3), they swapped the values of x and y.
7. Add the equations: 4x = 16, so x = 4. Then 4 - 2y = 0 gives y = 2. Check: 3 × 4 + 2 × 2 = 16.
8. Subtract the equations: 2 bags and honey minus 1 bag and honey leaves 1 bag, so 40 - 28 = AED 12. Then the honey is 28 - 12 = AED 16.
9. Replace x with 3: 2 × 9 - 3 × 3 + 1 = 18 - 9 + 1 = 10.
10. A. x^2 - 2x - 15. Multiply every term by every term: x^2 - 5x + 3x - 15 = x^2 - 2x - 15. If your child chose C (x^2 - 15), they multiplied only the first and last terms.
11. C. x = 2 or x = 3. Find two numbers that multiply to 6 and add to -5: (x - 2)(x - 3) = 0, so x = 2 or x = 3. If your child chose A (x = -2 or x = -3), they forgot the brackets hold minus signs, so the answers are positive.
12. Cups cost 3c plus 8 for the snack, and the total is at most 50: 3c + 8 <= 50. Then 3c <= 42, so c <= 14: she can buy at most 14 cups.
13. Maryam: 200 + 4 × 50 = AED 400 (linear, adds 50 each time). Hamdan: 100 × 2^4 = AED 1,600 (exponential, doubles each time), so Hamdan has more.

Section 2: Shapes, measuring and space

1. A. 10 units. The changes are 6 and 8: 36 + 64 = 100, and the square root of 100 is 10. If your child chose C (100 units), they forgot to take the square root.
2. C. (2, 7) Average the coordinates: (-2 + 6) / 2 = 2 and (4 + 10) / 2 = 7. If your child chose A (4, 14), they added the coordinates but did not halve them.
3. The midpoint is the same distance from both stations. From 1 to 3 is +2, so 3 + 2 = 5, and from 2 to 5 is +3, so 5 + 3 = 8. The other station is at (5, 8).
4. B. 2. Rise 11 - 3 = 8 over run 6 - 2 = 4, so the slope is 8 / 4 = 2. If your child chose D (8), they used the rise and forgot to divide by the run.
5. Parallel lines have the same slope, 2, and this line crosses the y-axis at (0, 5), so y = 2x + 5.
6. D. -1/3. Flip 3 to 1/3 and change the sign to get the negative reciprocal, -1/3. If your child chose B (-3), they changed the sign but forgot to flip.
7. Yes. Multiply the slopes: 2/3 x (-3/2) = -1, and slopes that multiply to -1 mean perpendicular lines.
8. The perpendicular slope to -1/2 is 2. Use y - 4 = 2(x - 1), so y = 2x - 2 + 4 = 2x + 2.

9. The short sides are 4 m and 3 m. The long side is the square root of 16 + 9 = 5 m, so the fence is 4 + 3 + 5 = 12 metres.
10. Volume = 22/7 x r squared x h = 22/7 × 49 × 10 = 22 × 7 × 10 = 1,540 cubic m.
11. B. 376.8 cubic cm. A cone is one third of pi x r squared x h: 1/3 × 3.14 × 36 × 10 = 376.8 cubic cm. If your child chose A (1,130.4), they forgot the one third.
12. Ball: 4/3 x pi x 27 = 36 pi, about 113.04 cubic cm. Doubling the radius gives 2 cubed = 8, so the volume is 8 times as large.

Section 3: Data, graphs and chance

1. B. 27. Add the students who walk: 12 + 15 = 27. If your child chose C (30), they counted a whole group of 30 instead of only the walkers.
2. Girls: 12 out of 30 is 12 / 30 = 40%. Boys: 15 out of 30 is 15 / 30 = 50%.
3. C. 25%. Boys who take the bus are 15 out of all 60 students, so 15 / 60 = 25%. If your child chose B (50%), they used only the 30 boys as the total.
4. Put x = 4 into the line: 5 × 4 + 50 = 20 + 50 = 70.
5. Slope 5: each extra hour of study goes with about 5 more points. Intercept 50: with 0 hours of study the predicted score is 50.
6. A. 4. Residual = actual minus predicted = 74 - 70 = 4. If your child chose B (-4), they subtracted the other way round.
7. D. r = -0.8. Negative means below 0 and strongest means closest to -1, so r = -0.8. If your child chose A (r = 0.9), they picked a strong relationship, but it is positive.
8. No. Age explains both, since older children have bigger feet and more reading practice. Correlation does not mean causation.
9. C. 5. In order, the lower quartile is 4 and the upper quartile is 9, so the IQR is 9 - 4 = 5. If your child chose A (10), they found the range, 12 - 2, instead.
10. The five numbers are the minimum 2, lower quartile 4, median 7, upper quartile 9 and maximum 12. The box runs from 4 to 9.
11. Before: mean (5 + 6 + 7 + 8 + 9) / 5 = 7 and median 7. After: mean 75 / 6 = 12.5 and median (7 + 8) / 2 = 7.5. The median is better because 40 pulls the mean up a lot.

Section 4: Numbers and calculating

1. A. 2^8. Same base, so add the exponents: 2^(5 + 3) = 2^8 = 256. If your child chose B (2^15), they multiplied the exponents instead of adding.
2. Same base, so subtract the exponents: 5^(6 - 4) = 5^2 = 25.
3. C. 64. The square root of 16 is 4, and 4 cubed is 4 × 4 × 4 = 64. If your child chose A (24), they multiplied 16 by 3/2 instead of using it as a power.
4. The cube root is the power 1/3, so the cube root of 5 is written 5^(1/3).
5. A. 1/25. A negative exponent means the reciprocal: 5^-2 = 1 / 5^2 = 1/25. If your child chose B (-25), they made the answer negative instead of taking the reciprocal.
6. B. 3 + the square root of 2. is irrational, because 2 is not a perfect square. If your child chose C (the square root of 16), they saw a root sign, but it equals 4, which is rational.
7. Rational: the square root of 2 x the square root of 8 = the square root of 16 = 4, a whole number.
8. 108 km is 108,000 m and 1 hour is 3,600 s, so 108,000 / 3,600 = 30 metres per second.
9. 140 km is 1.4 times 100 km, so the fuel is 8 × 1.4 = 11.2 litres.

10. C. 4.5 × 10^7. Move the decimal point 7 places left to get 4.5, so 45,000,000 = 4.5 × 10^7. If your child chose A (4.5 × 10^6), they counted one place too few.
11. Multiply: 2 × 3 = 6 and 10^4 × 10^3 = 10^7, so 6 × 10^7. Divide: 6 / 3 = 2 and 10^8 / 10^5 = 10^3, so 2 × 10^3.
12. 3 × 10^6 × 7 = 21 × 10^6 = 2.1 × 10^7 litres. Cubic metres are an easier unit: 21,000 cubic metres.